

ZAID AHMAD KHAN

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SUMMARY

Mechanical and Aerospace Graduate student specializing in robotics prototyping, CAD, and deep reinforcement learning (DRL). Built and tested 10+ 1:14 scale autonomous vehicles for student cohorts at MIT BWSI. Experimental testing using industry-grade equipment like waterjet, 3D printer, wind tunnel, beam-bending rigs, etc. with DFM and DFA practices.

EDUCATION

Illinois Institute of Technology, Chicago, IL

May 2027

MS in Mechanical & Aerospace Engineering, GPA: 3.8/4.0

- Researched Reduced-Order Modeling (ROM) methods in turbulence modeling and Neural Operators (DeepONet) for the prediction of nonlinear fluid dynamics.
- Implemented Data-Driven Techniques to predict RUL of aging Li-Ion batteries. Datasets from NASA & Kaggle.
- Selected as Honor Marshal for the Commencement Ceremony; Awarded 40% merit-based tuition scholarship.

University of Salford, Manchester

June 2024

Beng (Hons) in Aeronautical Engineering, CGPA: 3.83/4.0 (First Class Honors)

- Researched aircraft wing and jet dynamics through CFD (Ansys Fluent) and wind tunnel experimentation.
- Conducted laboratory tests in structural analysis of beams, PID controls, aerodynamics, material testing, etc.
- Ranked 2nd in the major and awarded the Gold Excellence Scholarship for academic performance.

WORK EXPERIENCE

Autonomous RACECAR Associate Instructor / TA

MIT Lincoln Laboratory / BWSI, Cambridge, MA

July 2026 - current

- Upgraded, fabricated, and maintained a fleet of 15 physical RACECAR Neo units for student engineering cohorts in a lab-based environment; prototype testing through dev car.
- Manufactured custom PCBs, power systems, and chassis components through precision-soldering of wires, header pins, and copper strips.
- 3D printed custom structural mounts, routed power and data cables, and implemented secured bolted, screwed, and magnetic attachments.
- Co-delivered lectures and guided hands-on labs in advanced autonomy concepts, including classical/modern control theory (PID, LQR, MPC), sensor fusion (EKFs), computer vision, and deep reinforcement learning.

VOLUNTEER EXPERIENCE

Graduate Student Consultant

The Build Fellowship / Open Avenues, Chicago, IL

February 2026 - April 2026

- Designed and modeled a battery enclosure layout for an EV (Ford Mach-E) to safely pack a 160L battery payload using Metal Sheet workspace in Fusion 360 complying with AHSS & DFM.
- Calculated AHSS-based spring back factor ($K_s = 0.984$), cold-forming parameters, tool die angles (81.28 degrees) and verified analytical models against physical constraints.
- Conducted FEA stress simulations for underfloor intrusion, and joint integrity to ensure rigorous FMVSS 305 compliance.
- Drafted comprehensive GD&T drawings, Bills of Materials (BOM), joint analyses, and hand stress calculations.
- Accounted for corrosion protection through resistance spot-welding techniques (multi-pulse weld cycles) and liquid metal embrittlement.

Deep Reinforcement Learning Research Volunteer

Neobotics Foundation Inc, Boston, MA

February 2026 – April 2026

- Architected RL model using Unity ML-Agents and Proximal Policy Optimization, RC car completes L-shaped track within 12 seconds. [<https://github.com/Neobotics-Foundation-Inc/NeoRacer-RL>]
- Engineered a dense, continuous reward function utilizing Power Mean smoothing and leveraged Behavioral Cloning to bypass random exploration and stabilize initial neural network training.
- Authored comprehensive curriculum materials, including an RL Architecture Report and a hands-on Student Training Guide.

Technical Project Director

American Institute of Aeronautics & Astronautics (AIAA), Illinois Tech, Chicago, IL

Sep 2025 - current

- Presided over 10+ technical executive board meetings to manage programmatic timelines, project scopes, and computing resources for student engineering tracks using Microsoft Teams, Forms, Slack, and GroupMe.
- Formulated and instructed weekly advanced CAD and engineering design workshops using Fusion 360 to train a cohort of 30+ student members.
- Taught advanced modeling techniques by designing complete elliptical wing structures, including struts, spars, ribs, flaps, and slats.

Waterjet Technician

Salford Racing Team, Manchester, UK

Jan 2023 – July 2023

- Operated waterjet cutting systems to fabricate 30+ metal and carbon-fiber components for F1 chassis, exhaust, and structural assemblies, in 100% compliance with detailed engineering drawings and safety procedures.
- Organized 20+ part drawings using Adobe Illustrator and specifications to support accurate, repeatable fabrication and on-time delivery.
- Designed and constructed two custom shadow boards to optimize workshop resource management and improve workflow efficiency by 15%.
- Maintained strict dimensional machining tolerances throughout the fabrication process, in accordance with DFM.

EXPERIMENTAL PROJECTS & RESEARCH

Transonic Wing Prototyping & Wind Tunnel Validation | SolidWorks, CNC, Excel

January 2024 – April 2024

- Modeled a 0.5-meter span 3D wing structure in SolidWorks by lofting two distinct transonic airfoils at the root and tip.
- Engineered a robust dovetail joint to physically assemble the scaled model, effectively overcoming standard 3D printer volume limits.
- Introduced 10 surface pressure taps to collect precise pressure variations over and underneath the wing while complying with manufacturing constraints.
- Executed subsonic wind tunnel testing between flight speeds of 5 to 25 m/s to capture divergence Mach numbers and identify boundary layer separation points, validating design.

Design and Assembly of Surface Gauge | SolidWorks

January 2022 – April 2022

- Modeled individual assembly parts for a surface gauge and drafted technical drawing using SolidWorks.

Manufacturing of Screw Gauge | Industry Tools

October 2023

- Manufactured a screw gauge from mild steel using hand tools and machinery as part of a practical workshop at Trafford College, Manchester.
- Implemented GD&T principles to understand technical blueprint of the design and manufacture accurately.

SKILLS

- **CAD, FEA:** SolidWorks, Ansys Mechanical APDL, Fusion 360, AutoCAD.
- **Hardware:** PCB Fabrication, Soldering (Paste/Iron), 3D Printing, Hardware Debugging, DFA, DFM, Waterjet CNC
- **Data Science:** Python, PyTorch, Unity ML-Agents, NumPy, Matplotlib, Conda, VS Code, Git/GitHub, Linux WSL.
- **CFD:** Ansys Fluent, OpenFOAM, SimScale, Xfoil, Turbulence Modeling (RANS, Spalart-Allmaras).

